



Deffy

Site inspection report in seconds not hours

AI Hackathon 2025

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Team Members:

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TOP Site Inspection Reports are manual and time consuming



Delayed rectification affects TOP timeline

 **Clunky tool that slows down workflows**

Compliance clause lookup is time consuming

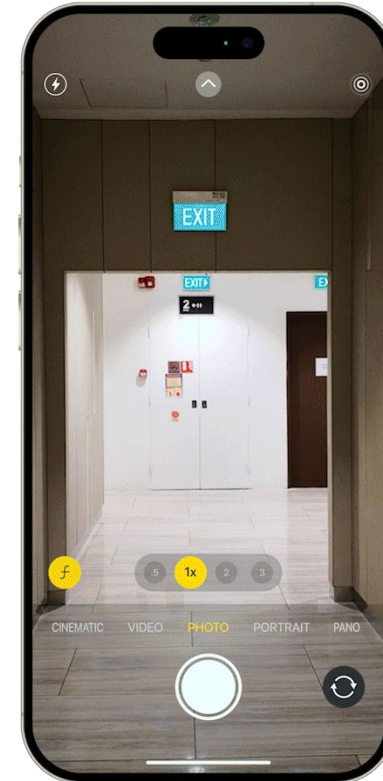
Meet Deffy - Your AI Agent for Site Inspection

Getting Started is as easy as 1,2,3:

1. Find Deffy on Telegram (@AgentDeffyBot)
2. Send the defect image to Deffy
3. Review report on Excel

Simple UI/UX of a messaging app that is familiar to everyone.

Let AI do the mundane work of writing the report.



Scope and Objectives



Send Image of Defects



Verify Generated Report



Timely Issue Rectification

Project on time =  **Happy stakeholders**

Using Agentic AI, Deffy helps to **automate report generation**

Objectives:

- Faster issue rectification
- Accurate clause referencing
- Improve and standardise report quality

Architects can **focus** on the **coordinations and issue resolution** rather than the manual paperwork.

Contractors will have **more time for rectification.**

Deffy AI will be a companion for young architects to **improve their knowledge** on codes.



Why AI is effective to reduce workload?

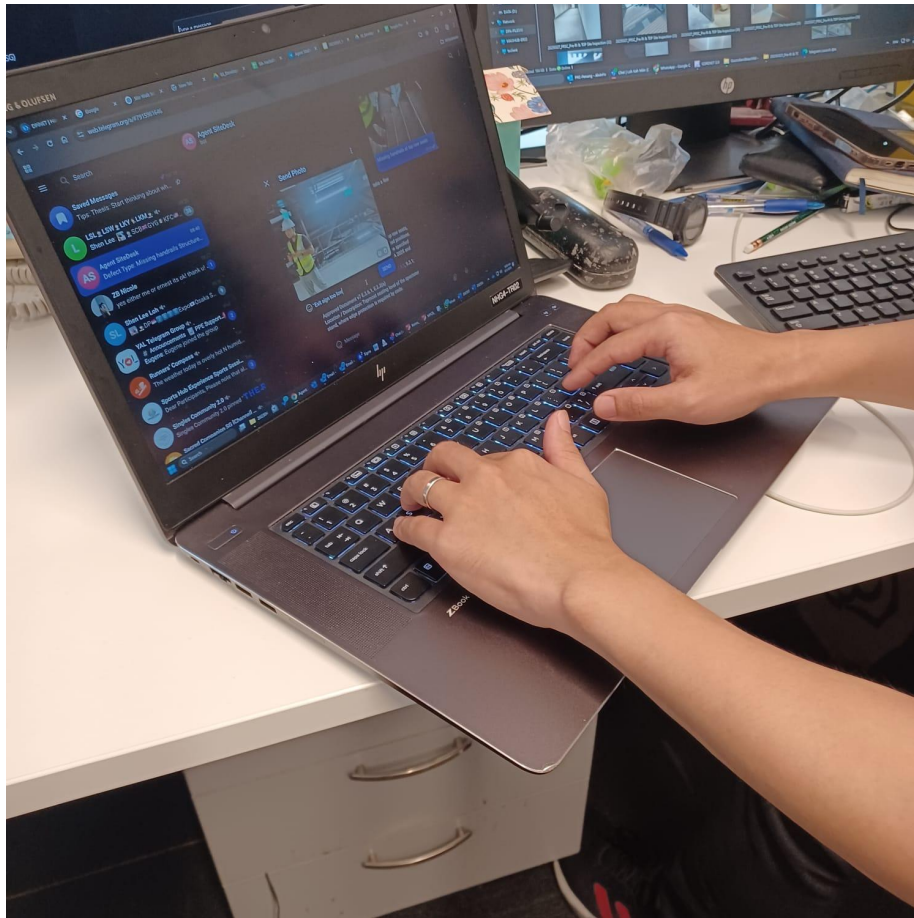
Deffy AI

- ✓ **Zero** learning curve
- ✓ **Simple UI** and uses natural language
- ✓ Fits into **existing workflow** and customisable reports

Legacy Tools

- ✗ 3 to 6 months of onboarding and **training**
- ✗ **Clunky forms** and complicated user interface
- ✗ Predefined workflow and report that **requires significant effort** for customisation

“5x faster than my current process”



User Testimonial

“Less time is spent on manually inputting information, I now have **more time to vet for accuracy**”

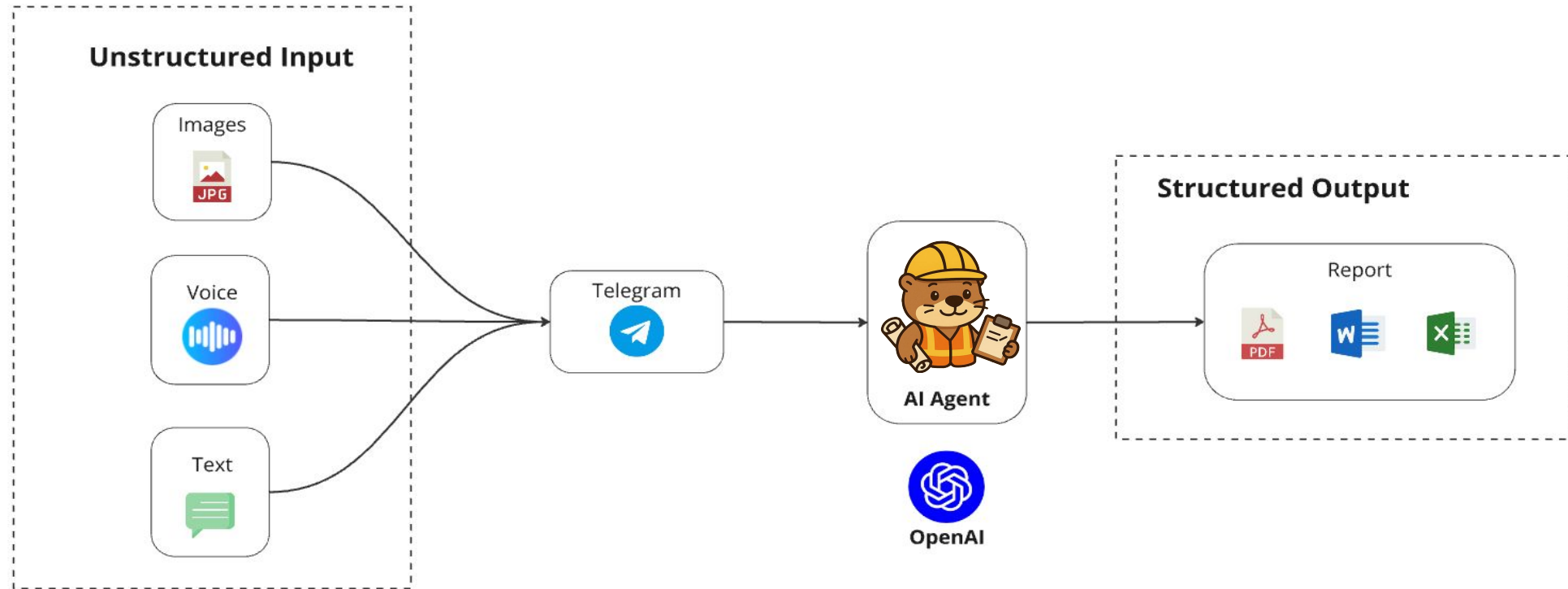
“It inserts the photos automatically into the table, so **it is fast**”

Acknowledgement:

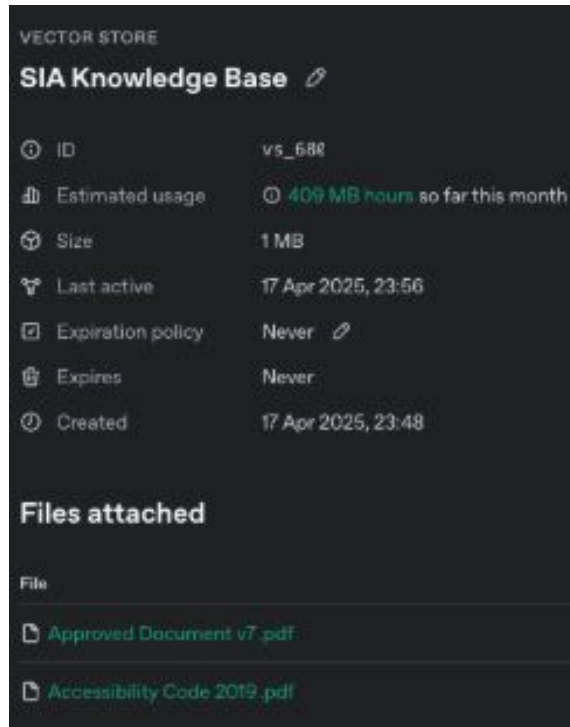
Photo of testing session with a practitioner from an architecture firm experienced in producing site inspection reports.

We sincerely thank him for his personal support in testing our prototype chatbot and for sharing valuable feedback based on his expertise.

Under The Hood - Multimodal Agentic Flow



Compliance Code Check using OpenAI RAG



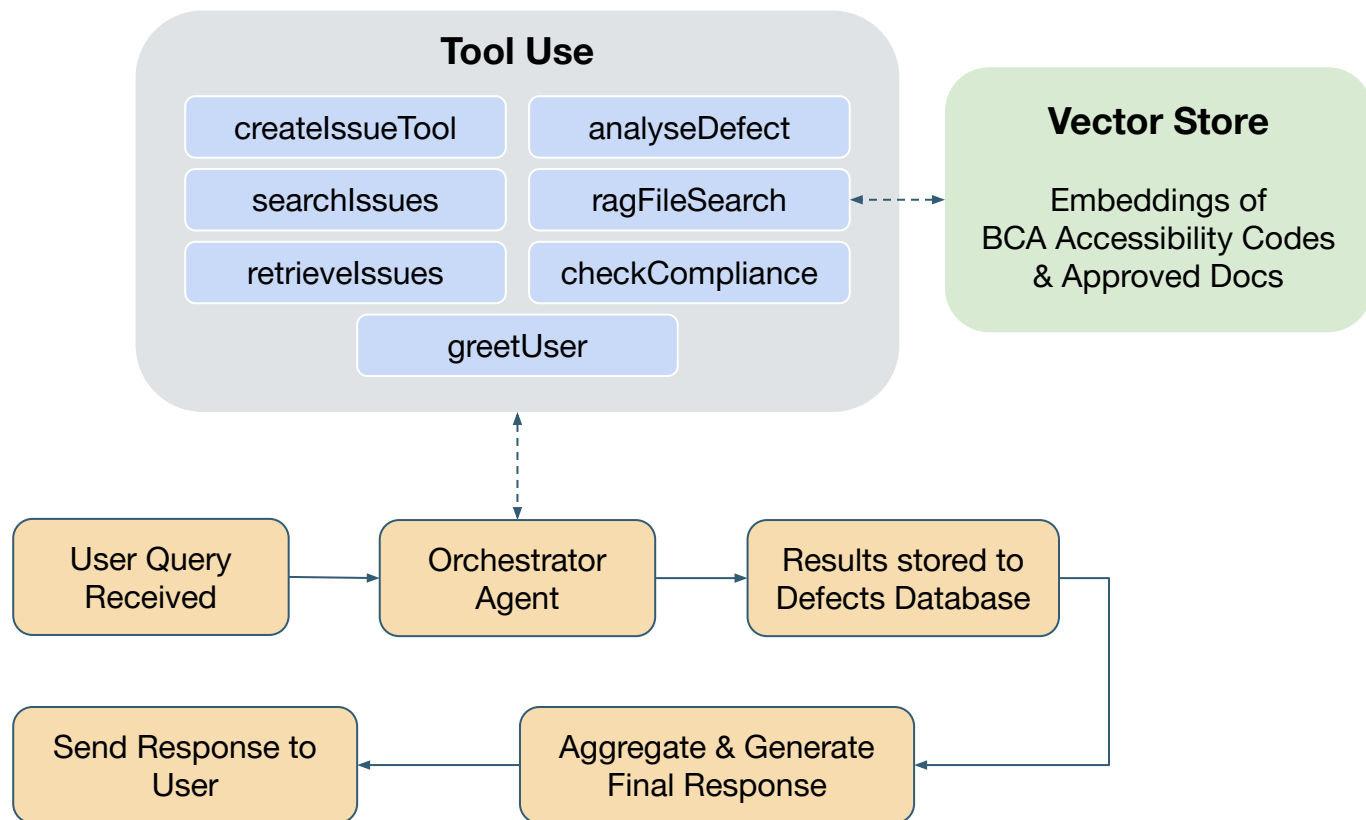
 Search through multiple compliance documents in seconds

 Evaluate defect based on relevant compliance clauses

 Suggest actions to proceed with rectification work

Deffy memorises the latest codes so you don't have to!

Agentic Flow - Tool Use Pattern



Agent Tools using OpenAI's gpt-4.1

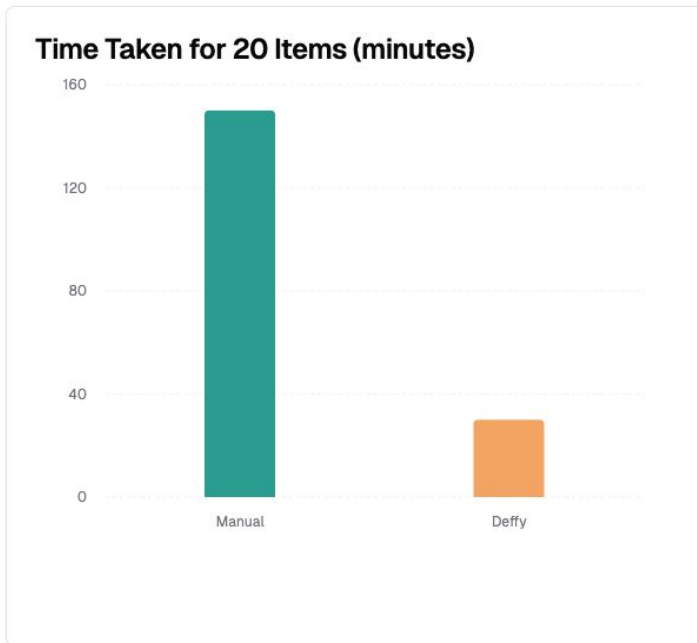
Why we use an AI Agent:

1. Large amounts of unstructured data
2. Difficult-to-maintain rules
3. Complex process to follow

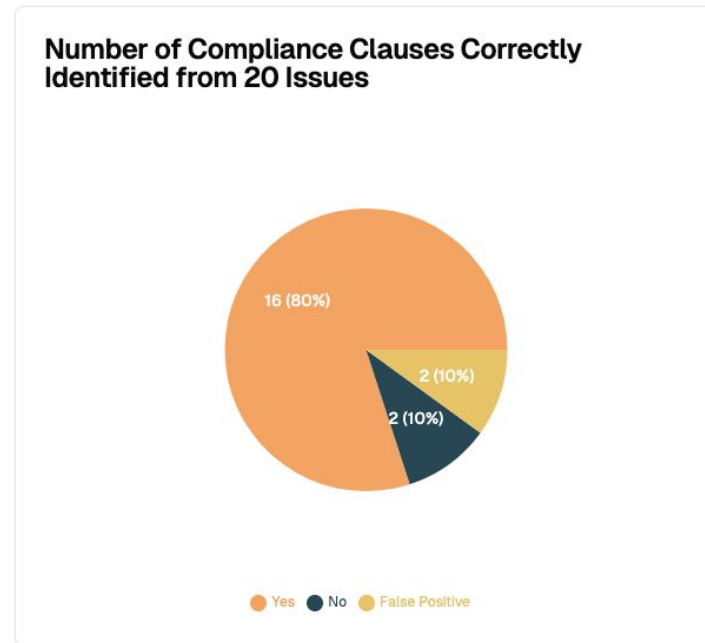
We breakdown the tasks into smaller steps with clear actions and defined outputs

5x faster at 80% of accuracy

5x faster than conventional process
(150 minutes vs 30 minutes)



80% accuracy identifying correct clauses



 **Time Saved =  Earned!**



Deffy AI

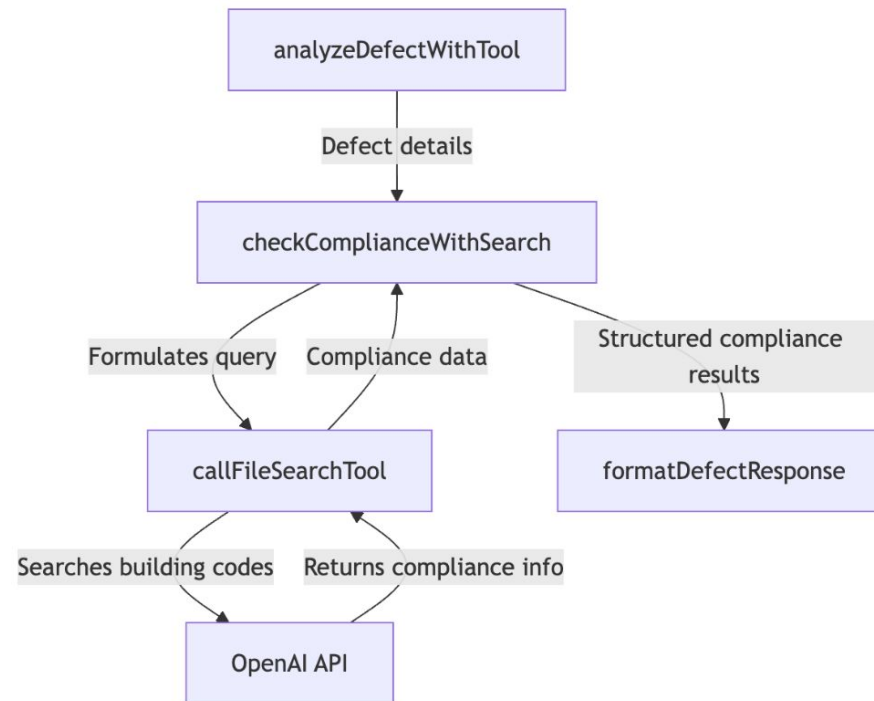
RAG Search Capabilities

OpenAI Tool Use and Function Registry

Compliance Code Checking

Integration with Defect Analysis

The RAG search capability is integrated with the defect analysis pipeline, particularly for compliance checking. The `checkComplianceWithSearch` function uses RAG search to determine if a detected defect violates any building codes or regulations.



Powered by GitBook

Direct RAG Search Usage

<https://wentilabs.gitbook.io>



Deffy AI

What's next?

Expansion to more compliance code

SCDF Fire code, NEA COPEH, etc to create a more comprehensive knowledge base to assist our users.

Linking plan drawings information

Allow location tracking corresponds to grid lines in plan drawings when user is on site so that issue can be tagged on the drawing.

Rectification work tracking

Allow contractor to provide issue updates so that everything can be tracked and dashboard to track progress of all issues.

Give us a try **TODAY!!!**

<https://t.me/AgentDeffyBot>



Appendix

Agentic Flow - Tool Use Pattern

```
// --- TOOL DEFINITIONS & HANDLERS ---

const analyzeDefect = {
};

const detectIntent = {
};

const retrieveIssues = {
};

const greetUser = {
  name: "greetUser",
  description: "Sends a welcome message to the user",
  parameters: {
    type: "object",
    properties: {
      customMessage: {
        type: "string",
        description: "Custom welcome message (optional)"
      }
    }
  },
  function: async (args) => {
  }
};

const checkCompliance = {
};

const searchIssuesTool = {
};

const createIssueTool = {
};

const ragFileSearch = {
  name: "ragFileSearch",
  description: "Searches BCA Code of Practice knowledge base for compliance information related to a defect",
  parameters: {
    type: "object",
  }
};
```

Properties	
Created	8 May 2025, 19:17
ID	chatcmpl-BUtXlzBB5Bv...
Model	gpt-4.1-2025-04-14
Tokens	888 total
Tools	analyzeDefect() detectIntent() retrieveIssues() greetUser() checkCompliance() ragFileSearch() searchIssues() createIssueTool()
Configuration	
Response	text

Defining an agent requires:

1. Model
2. Tools
3. Instructions & Prompts

Tools can be defined using few lines of codes

Prompt Engineering to achieve desired output

Before

Agent provides a long list of clauses without explanation

Broken Wall	Unknown	Unknown	Wall near meeting room area with TV screen	Inspect and assess	B.2.1, B.2.2, B.3.2.1 (Approved Document v7); 4.2.3.1, 4.2.3.2 (Accessibility Code 2019) - Approved Document v7.pdf B.2.1, B.2.2, B.3.2.1, Accessibility Code 2019.pdf 4.2.3.1, 4.2.3.2
Broken Wall	Unknown	Unknown	Meeting room (behind display screen and corner area)	Inspect and assess	B.2.1, B.2.2, B.3.1 (Approved Document v7.07); 4.2.3.1, 4.2.3.2 (Accessibility Code 2019); 9.1.1 (BCA COA 2025) - Approved Document v7.pdf B.2.1, B.2.2, B.3.1, Accessibility Code 2019.pdf 4.2.3.1, 4.2.3.2, BCA COA 2025.pdf 9.1.1
Grating gap greater than 12mm (measured at approximately 22mm)	Unknown	Unknown	floor grating area	Inspect and assess	4.1.2.1 (b) of the Code on Accessibility in the Built Environment 2019; 4.1.3 of the Code on Accessibility in the Built Environment 2025 - Accessibility Code 2019.pdf, Clause 4.1.2.1(b) and Figure 14, BCA COA 2025.pdf, Clause 4.1.3 and Figure 15

After

Agent provides only relevant clauses with and explanation to help practitioner quickly verify the clauses identified.

```
Notes: System , content: [ { type: 'input_text', text: 'You are an expert from Singapore's Building and Construction ' + 'Authority. Ignore all clauses from BCA COA 2025 or anything with 2025. Answer ONLY in JSON:\n' + '{\n  "clause": string,\n  "reasoning": string, \n  "citations": [string]]\n\n' + 'If no clause applies respond with "NA".' } ]
```

Missing toilet and nursing room signage; accessible toilet missing tactile and braille signage	Missing toilet and nursing room signage, as well as missing tactile and braille signage at accessible toilets, violates the requirement that public toilets must provide tactile and Braille signs. These signs must be mounted on the wall next to the latch-side of the door, at a specified height, and must comply with tactile and Braille specifications. This applies to office buildings, shopping complexes, multi-purpose complexes, sports complexes, hotels, boarding houses, serviced apartments, places of public resort, public swimming pools, markets, hawkers or food centres, transport stations, interchanges, passenger terminals, colleges, universities, institutions of learning, hostels, halls of residence or dormitories, hospitals and clinics.	Inspect and assess	5.1.8, 8.5.5, 8.5.4 - Accessibility Code 2019.pdf 5.1.8, 8.5.5, 8.5.4
Exit sign installed at an insufficient height above the mezzanine floor (too low)	Exit signs must be clearly visible to people who are walking, standing, or seated. The centre line of the sign must be at a height of 1500 mm above the floor level for tactile signs, and where signs would not be visible in a crowded situation, they should be suspended from the ceiling or projected from walls at a height of at least 2000 mm above the floor level. Installing an exit sign at an insufficient height above the mezzanine floor (too low) may violate these requirements if it is not clearly visible or does not meet the specified mounting heights.	Inspect and assess	8.5.4.3 and 8.5.4.4 of the Code on Accessibility 2019 - Accessibility Code 2019.pdf: 8.5.4.3, 8.5.4.4

Proper prompt engineering creates better response and more reliable output for the Deffy